Lo-fi Prototyping Report

Milestone 2

When Is Help Actually Helpful?

Rethinking Studying with AI

Group 3

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Declaration of Use of AI

We used Codex to improve grammar, wording and layout in the Lo-fi Prototyping Report. We also used it to assist with selected parts of the prototype and storyboard designs in Figma.

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Section 1: Lofi Prototypes

Idea A: A course-based study guide

Idea A helps students work through a topic using their own course material and explanations matched to their starting knowledge. The student connects to Moodle through an institutional sign-in flow, selects subjects and materials to import, and chooses a learning goal. A two-part knowledge check separates course foundations from knowledge of the selected topic. The results identify gaps and offer a prerequisites chat alongside the option to enter the topic chat. In the prerequisites area, the student can ask follow-up questions and take a separate check when ready. The topic chat breaks a proof into individually expandable steps so that the student can focus on the part they need. Page references open a course excerpt, while explanations added beyond the material are labelled separately. We want to test whether this structure makes help easier to follow, whether the initial checks feel worth the effort, and whether expanding steps leaves students actively reasoning rather than simply reading a solution.

Complete workflow. Idea A workflow file on GitLab. The full prototype also includes the learning-goal screen, prerequisites chat and test, a review-needed result, and all four proof steps. Figures 1–4 show key screens used in the storyboard.

Idea B: A student-led practice assistant

Idea B starts from the student's own attempt and offers more support only as it becomes necessary. The student describes their background, selects courses they have taken and opens a workspace with course files, a document preview and chat side by side. They add lecture material through a file picker or the drag-and-drop area, after which the prototype presents the topics found in the material. Instead of a quiz, students can rate their familiarity with each topic from 0 to 10 and add anything else they already know. They can ask a course question or choose a practice problem, submit their own attempt and receive feedback on what is right or still missing. If they remain stuck, they can request a hint, a concept explanation, a similar example and eventually a full solution. A confidence panel explains the displayed score and shows the supporting slide or indicates that no matching source was found. We want to test whether self-assessment is understandable, whether students keep working through their own attempts, and whether the confidence explanation encourages checking the material rather than accepting the score on its own.

Complete workflow. Idea B workflow file on GitLab. The complete workflow includes profile editing, self-assessment, practice and help escalation as well as source and confidence views. Figures 5–8 show selected interaction screens.

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Idea A: Course setup and starting knowledge

Course context and prior knowledge. Importing selected materials addresses Need 3, help grounded in the course. Checking foundations separately from the chosen topic addresses Need 2, help that matches what the student already knows.

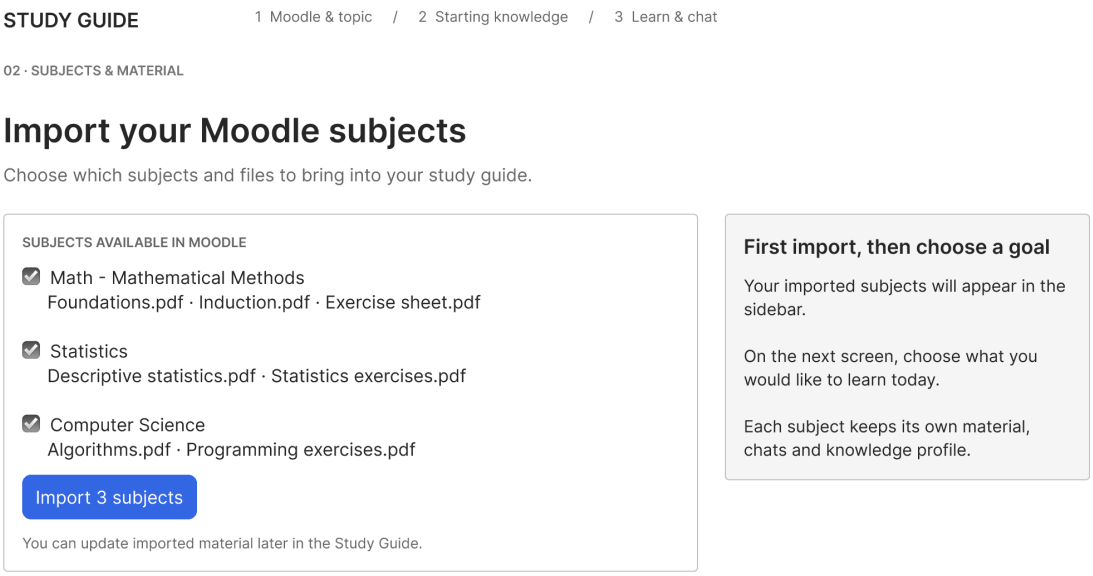


Figure 1: Select the course material. After the Moodle connection, the student chooses which subjects and files to import. The prototype represents the connection and import flow; it does not establish a working Moodle integration.

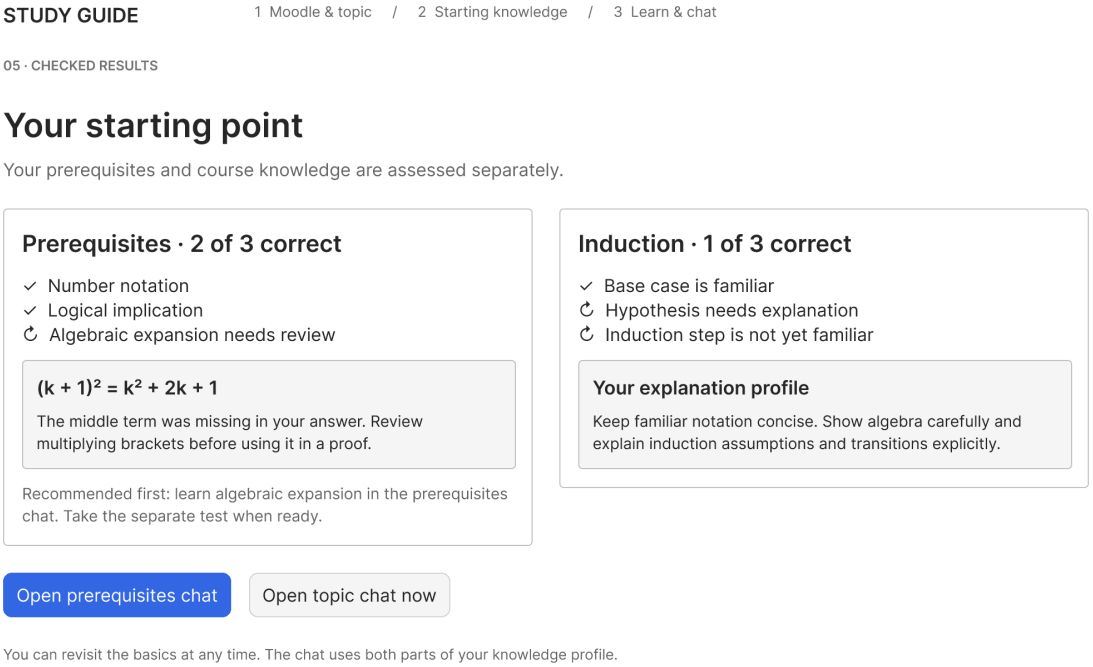


Figure 2: Review the starting point. Separate results for prerequisites and induction identify what needs explanation. The student can open the prerequisites chat or go to the topic chat.

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Idea A: Explore a step and inspect its source

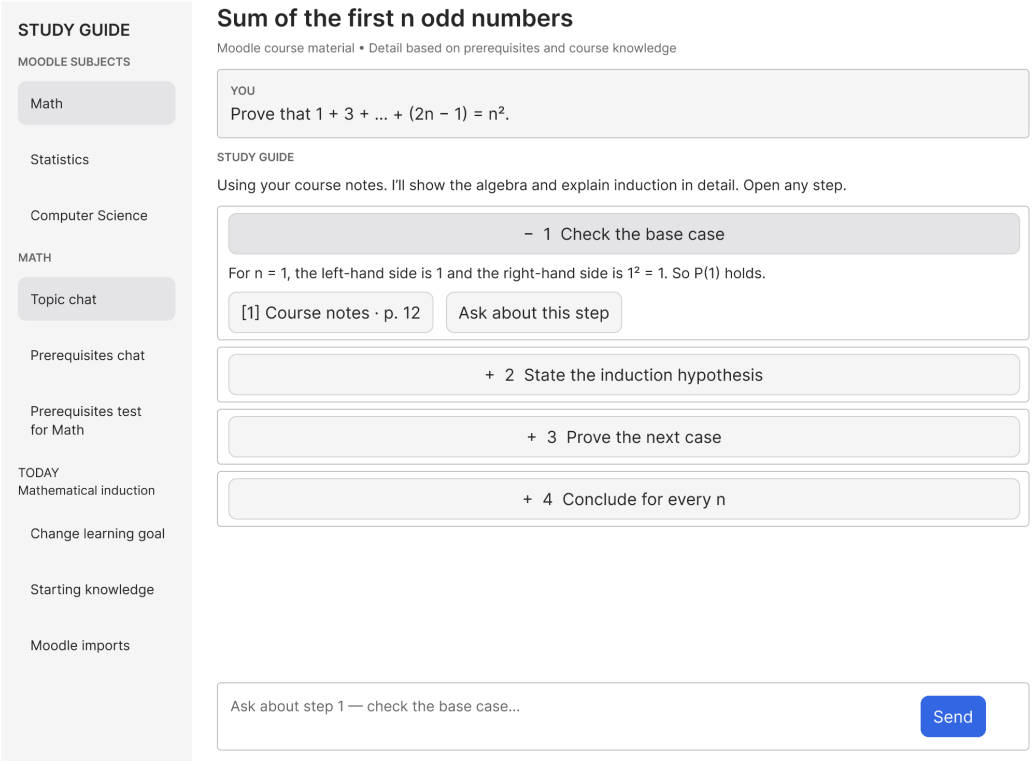


Figure 3: Open one proof step. The base case is expanded while the remaining steps stay collapsed. The student can ask about this step or open the cited page. This explores Need 1, short help with detail on request, and Need 4, keeping the student involved.

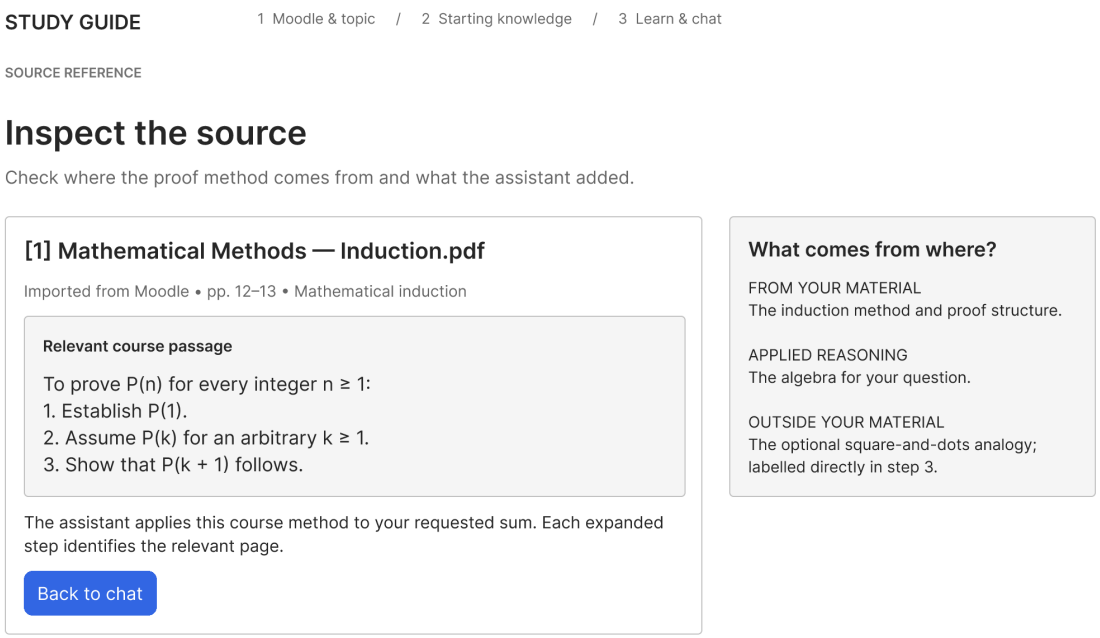


Figure 4: Check where the explanation comes from. The source view separates the course's induction method, reasoning applied to the exercise and an added analogy. It supports Needs 3 and 5 by making the basis of the answer inspectable.

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Idea B: Upload material and assess familiarity

A different starting point. Idea B replaces the Moodle connection with direct file upload and the initial quiz with a self-assessment. These are alternative ways to establish course context and starting knowledge, addressing Needs 2 and 3.

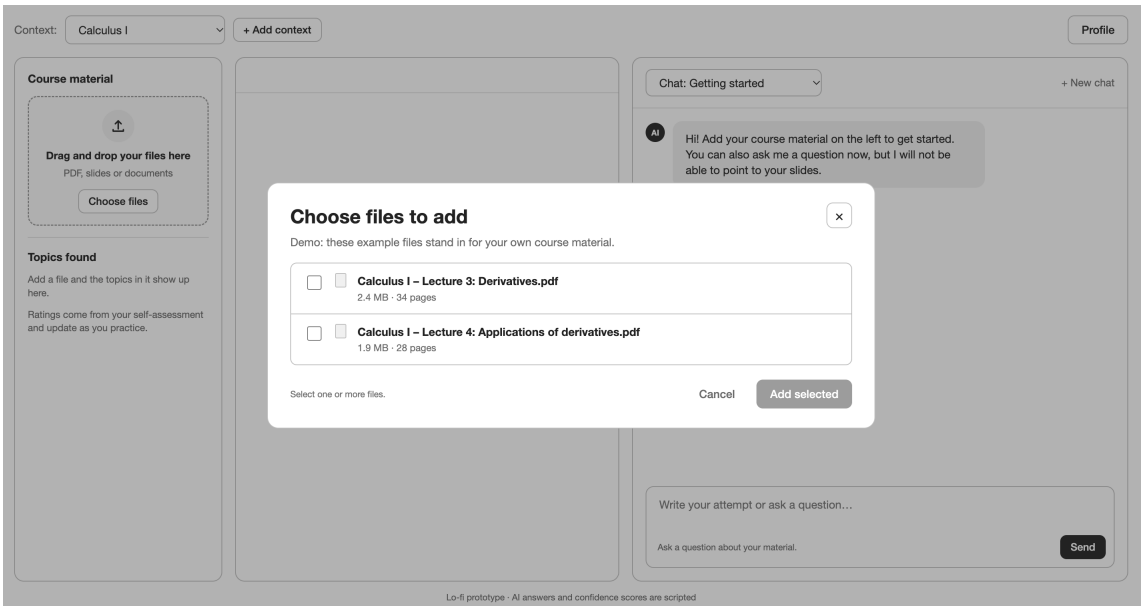


Figure 5: Add lecture files. The file-selection dialog allows the student to select material for the current course context. The example files simulate an upload; the workspace also presents a drag-and-drop area.

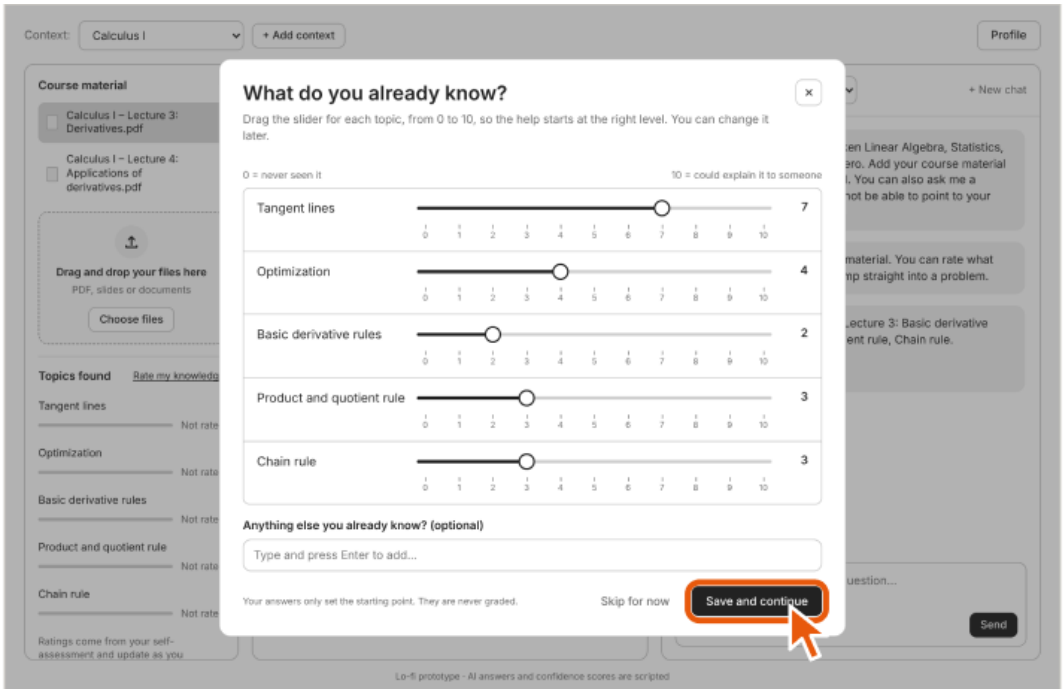


Figure 6: Rate what is already known. Students rate each topic from 0 (never seen it) to 10 (could explain it to someone). Ratings can be skipped or changed later. This screen is taken from the supplied interaction storyboard.

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Idea B: Keep context visible and judge the answer

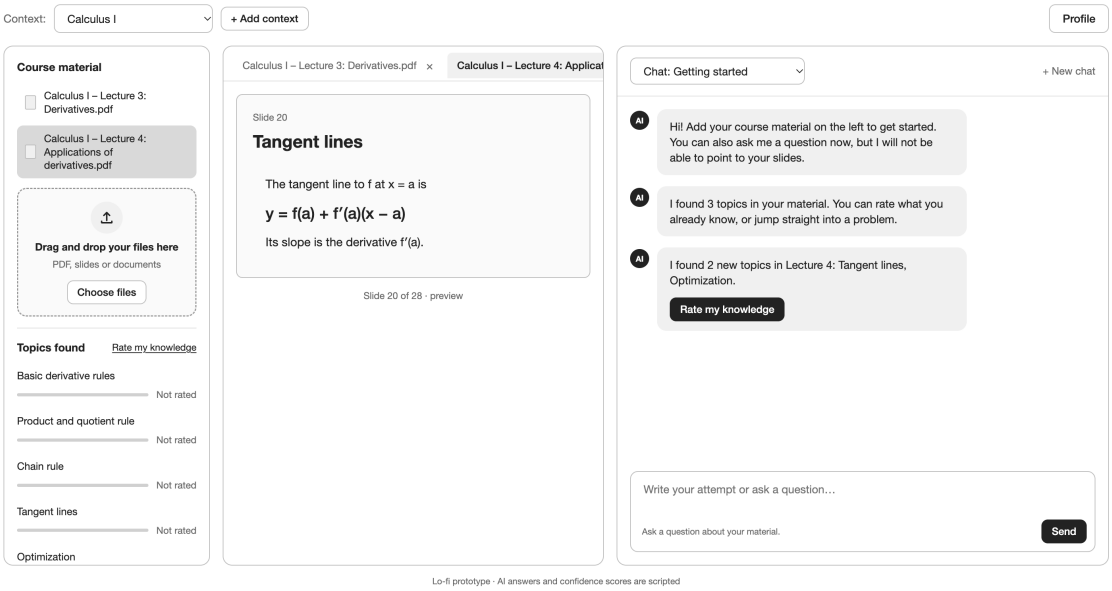


Figure 7: Work beside the course material. Files and topic ratings remain on the left, the selected lecture is in the centre, and the conversation is on the right. The student can consult the material while asking for help or working on an attempt.

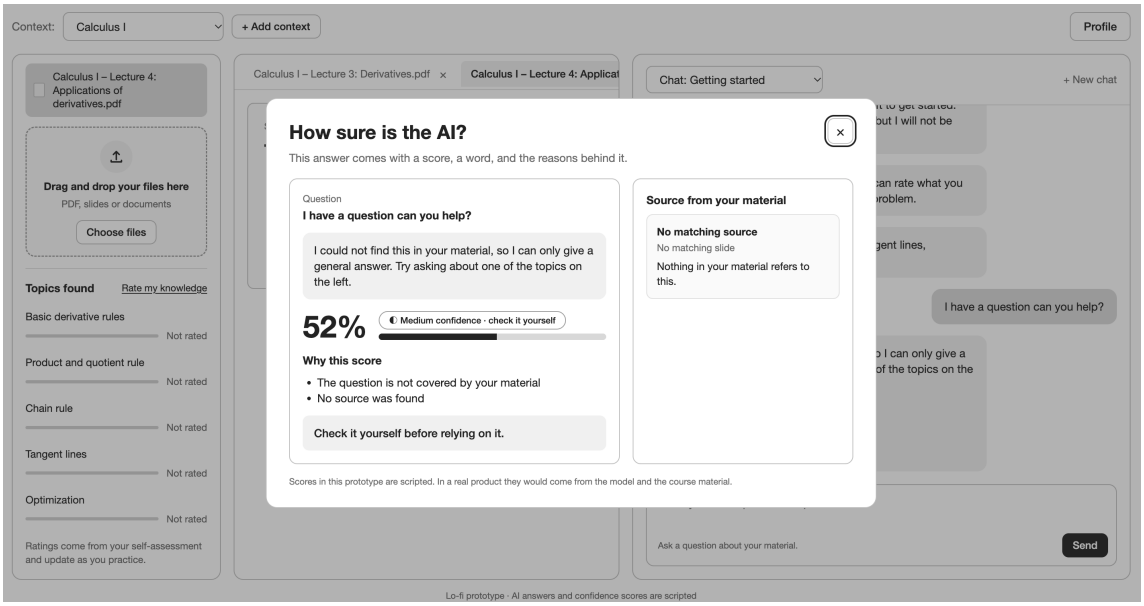


Figure 8: Inspect an answer without a matching source. The confidence dialog explains that the question was not covered by the uploaded material and that no source was found. The displayed 52% is scripted prototype content, not a measured probability that the answer is correct.

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Section 2: Interaction Storyboards

Storyboard – Idea A: From a knowledge gap to a course-based proof

User and goal. Maya, a first-year mathematics student, is preparing for a tutorial on her laptop in the library. She understands the base case but struggles with the algebra and induction step in $1 + 3 + \cdots + (2n - 1) = n^2$. Her goal is to explain the proof in her own words using her course notes. The original Figma storyboard is reproduced across Figures 9–12; read each page left to right, then the next row. Orange outlines mark the action in each screen.

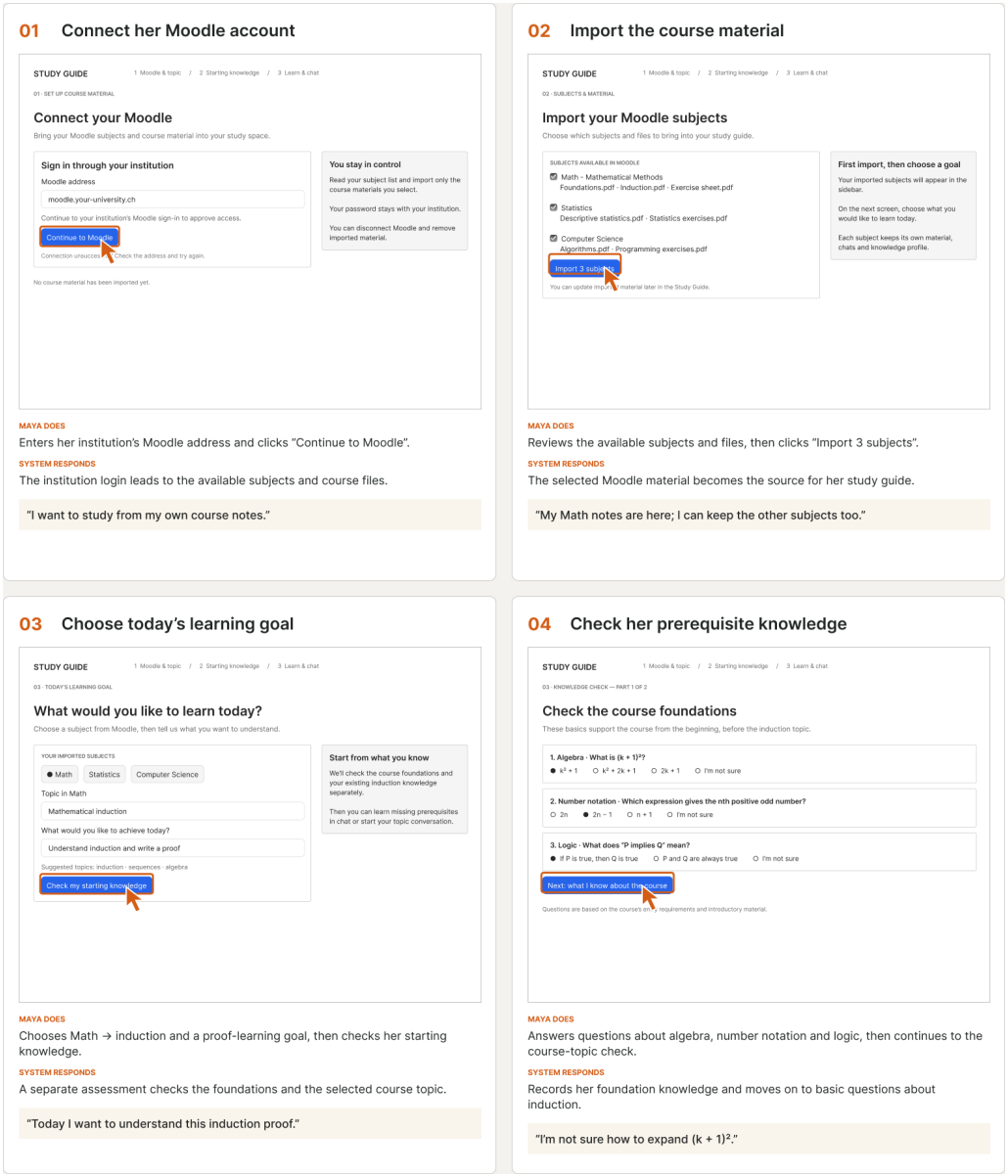


Figure 9: Storyboard – Idea A, panels 01–04. Maya connects Moodle, imports her material, selects induction as her goal and checks her knowledge of the course foundations.

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Storyboard – Idea A: Identify the gap and learn the missing foundations

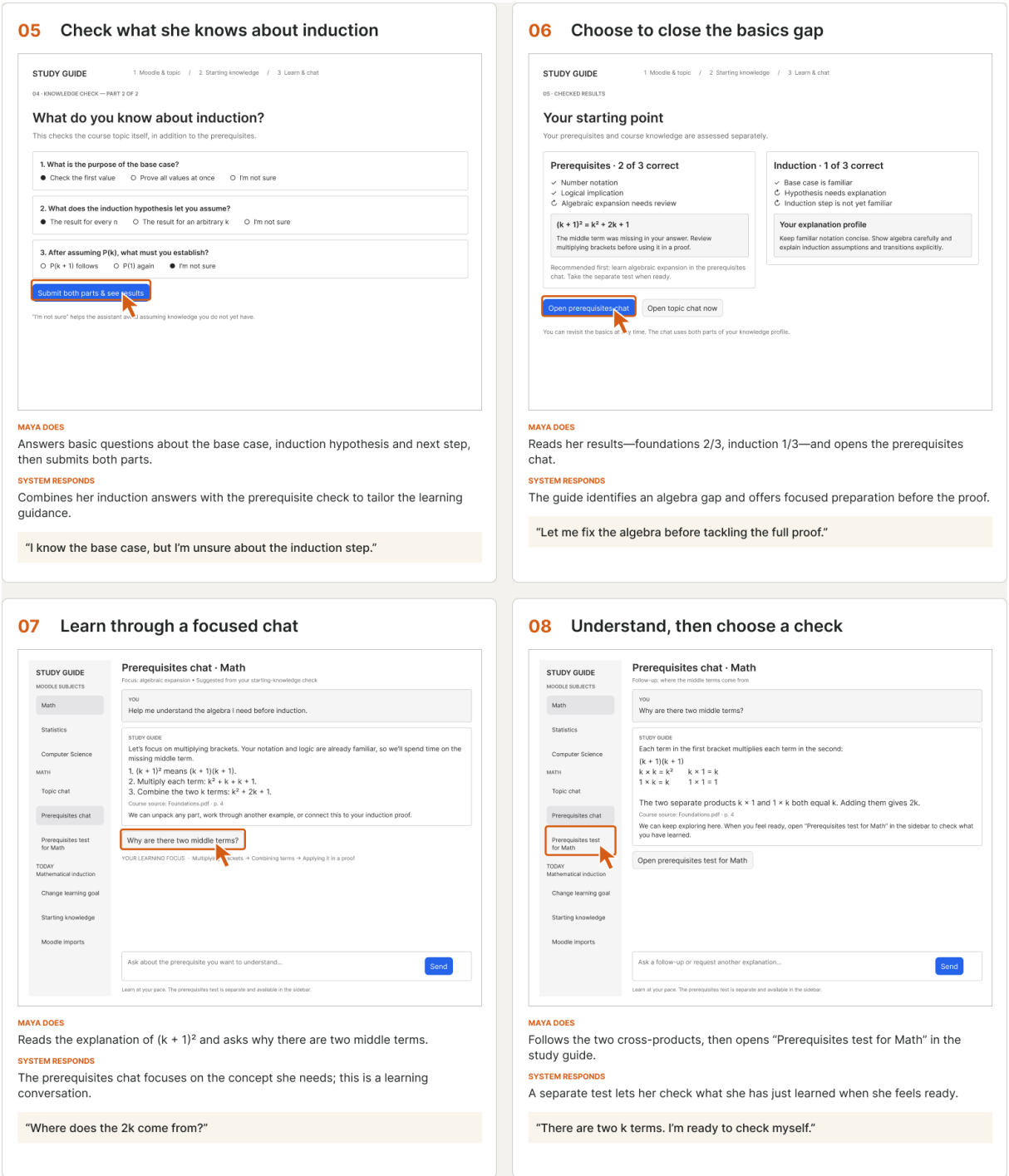


Figure 10: Storyboard – Idea A, panels 05–08. The two-part check reveals an algebra gap. Maya opens the prerequisites chat, asks why $(k+1)^2$ has two middle terms and follows the explanation before choosing to test her understanding.

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Storyboard – Idea A: Check understanding and return to the proof

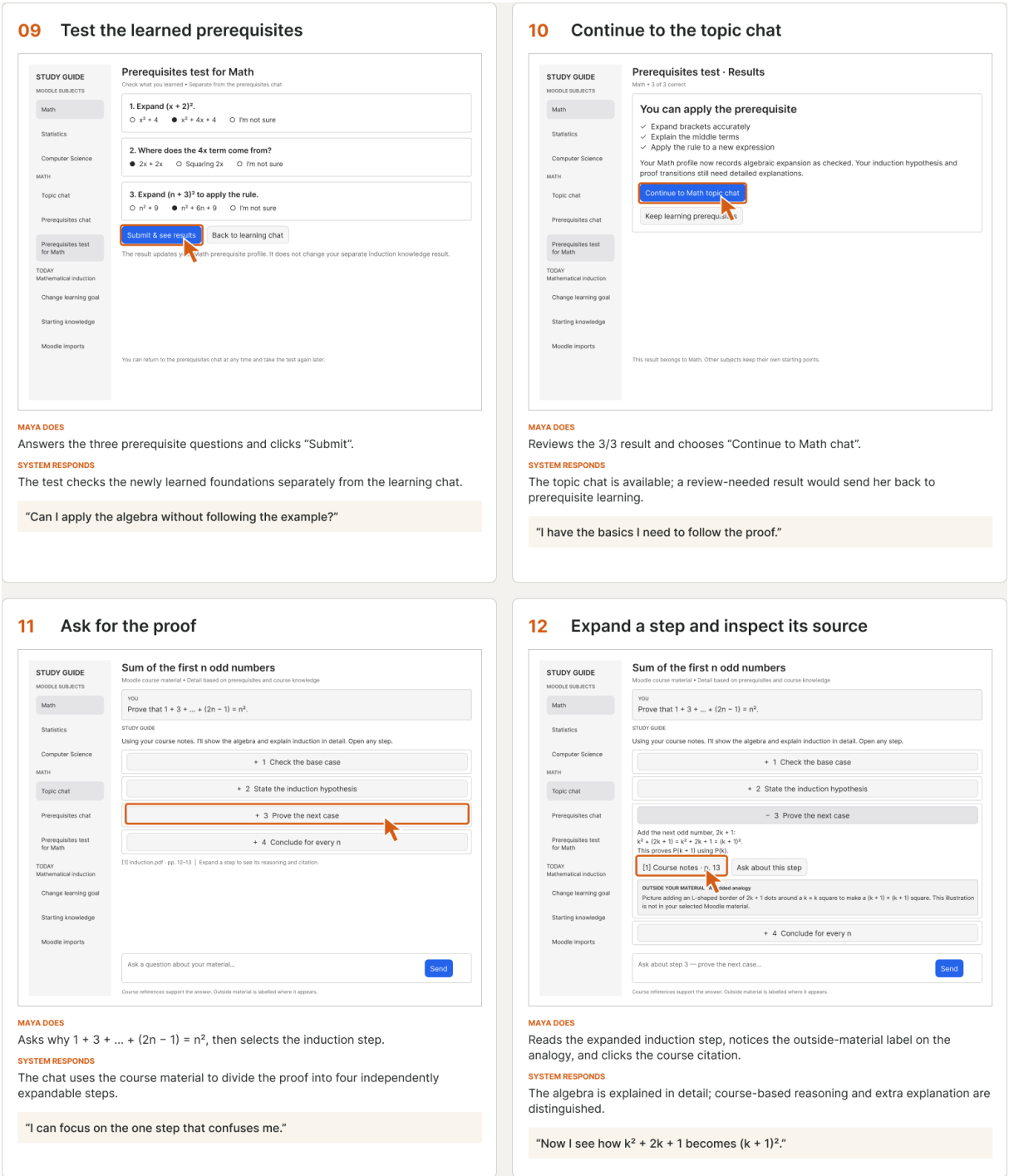


Figure 11: Storyboard – Idea A, panels 09–12. Maya takes the separate prerequisites test and returns to the topic chat. She requests the proof, opens the induction step and follows its course reference.

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Storyboard – Idea A: Verify the reasoning and reach the goal

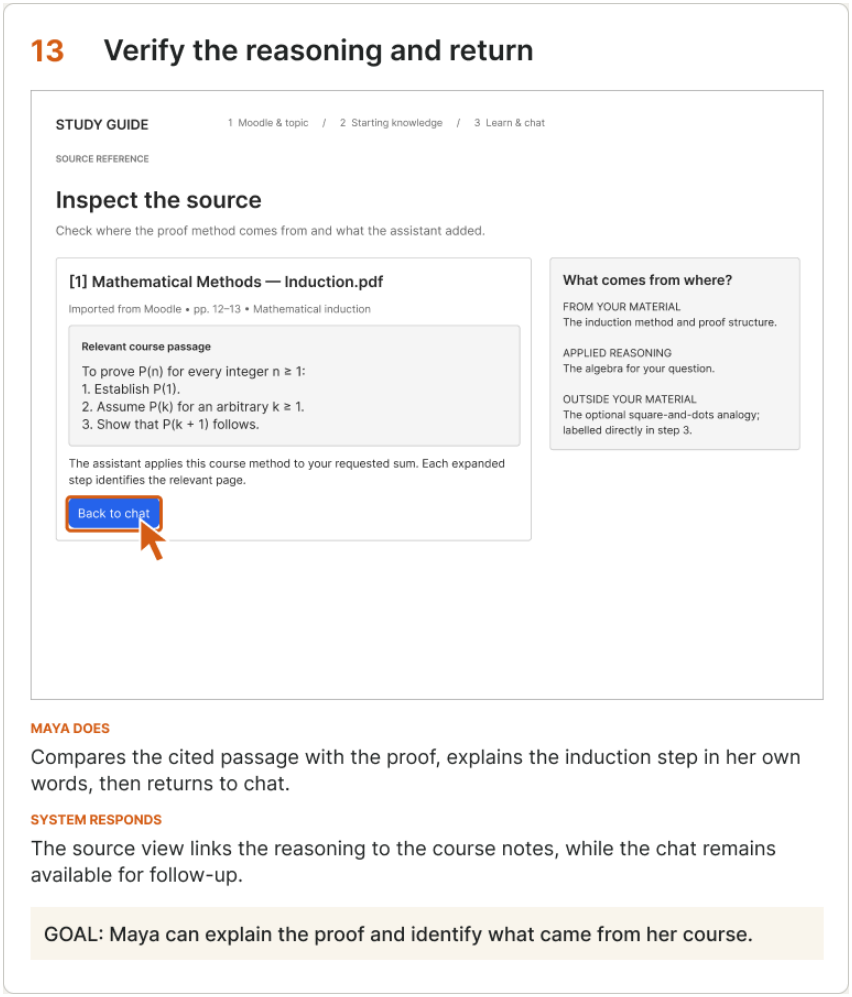


Figure 12: Storyboard – Idea A, panel 13. Maya compares the explanation with the cited course passage and returns to the chat.

How the task is accomplished. Maya first identifies and works on the algebra that prevents her from following induction. She then opens the proof step she needs and distinguishes the course method from the assistant's applied reasoning and additional analogy. The intended outcome is that she can explain why $k^2 + (2k + 1) = (k + 1)^2$ and identify the method in her notes.

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Storyboard – Idea B: From getting stuck to solving a problem independently

User and goal. Lena, a first-year student who has taken Linear Algebra and Statistics, is preparing a derivatives problem set on her laptop. She is unsure about the chain rule and wants to practise using her lecture slides, asking for only as much help as she needs. The original Figma storyboard is reproduced across Figures 13–15; read each page left to right, then the next row.

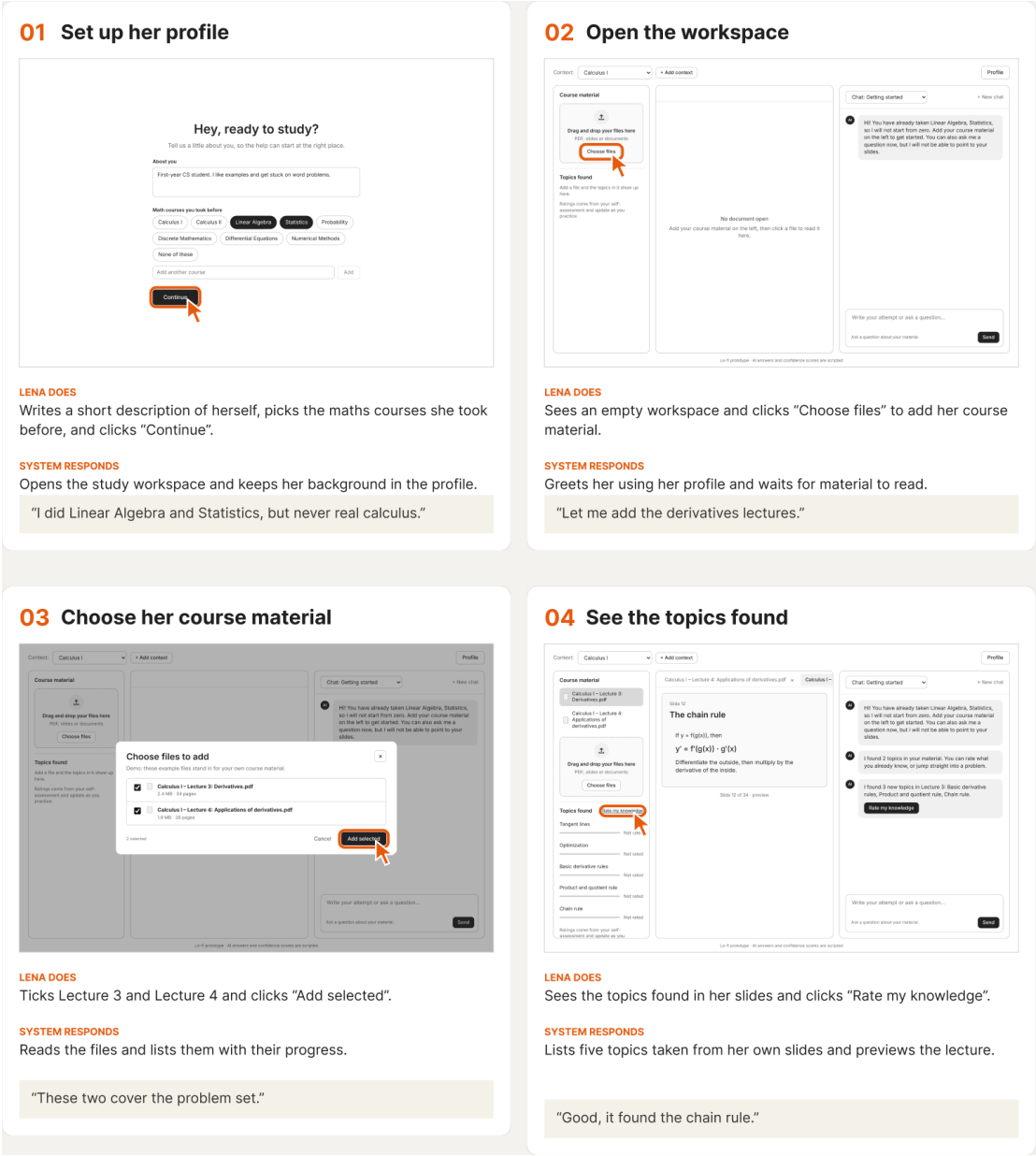
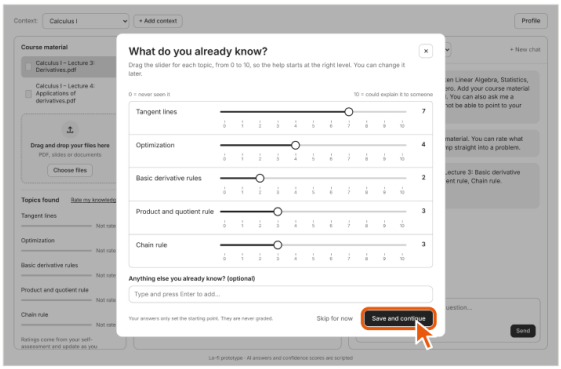


Figure 13: Storyboard – Idea B, panels 01–04. Lena sets up her profile, enters the workspace and adds two lectures. The system presents their topics and previews the material, ready for her to rate her knowledge or begin a question.

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Storyboard – Idea B: Assess familiarity, ask a question and check the source

05 Rate what she already knows

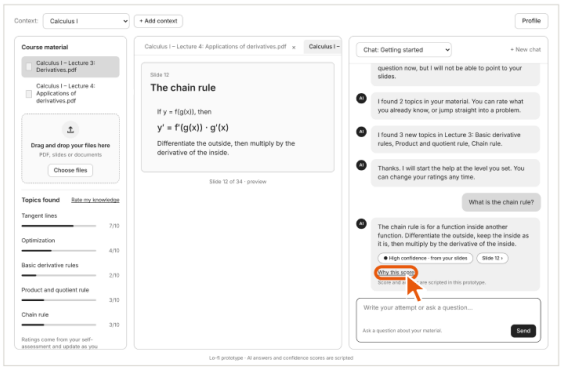


LENA DOES
Moves a 0 to 10 slider for each topic and clicks “Save and continue”.

SYSTEM RESPONDS
Saves her starting levels. They are never graded. Help starts from them.

> “Chain rule is my weak spot, about 2 out of 10.”

06 Ask a question about the material

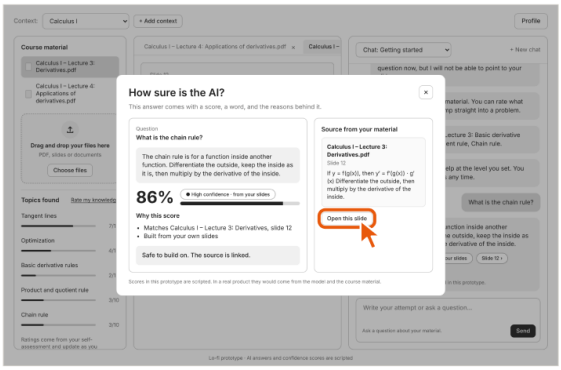


LENA DOES
Asks “What is the chain rule?” and clicks “Why this score” to check the answer.

SYSTEM RESPONDS
Answers from her slide and shows a confidence label with a link to the slide.

> “Is this from my slides, or general knowledge?”

07 Check the confidence and source

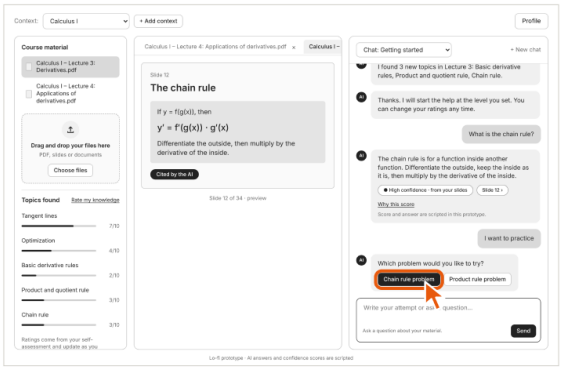


LENA DOES
Reads the score and its reasons, then clicks “Open this slide”.

SYSTEM RESPONDS
Shows the score for this answer, why it was given, and the exact slide text.

> “It matches my lecture, so I can build on it.”

08 Ask to practice



LENA DOES
Types “I want to practice” and clicks “Chain rule problem”.

SYSTEM RESPONDS
Offers problems for topics found in her material.

> “Let me try one myself first.”

Figure 14: Storyboard – Idea B, panels 05–08. Lena rates her familiarity, asks about the chain rule and inspects the confidence score, its reasons and the supporting slide. She then asks to practise and selects a chain-rule problem.

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Storyboard – Idea B: Attempt, request support and try again

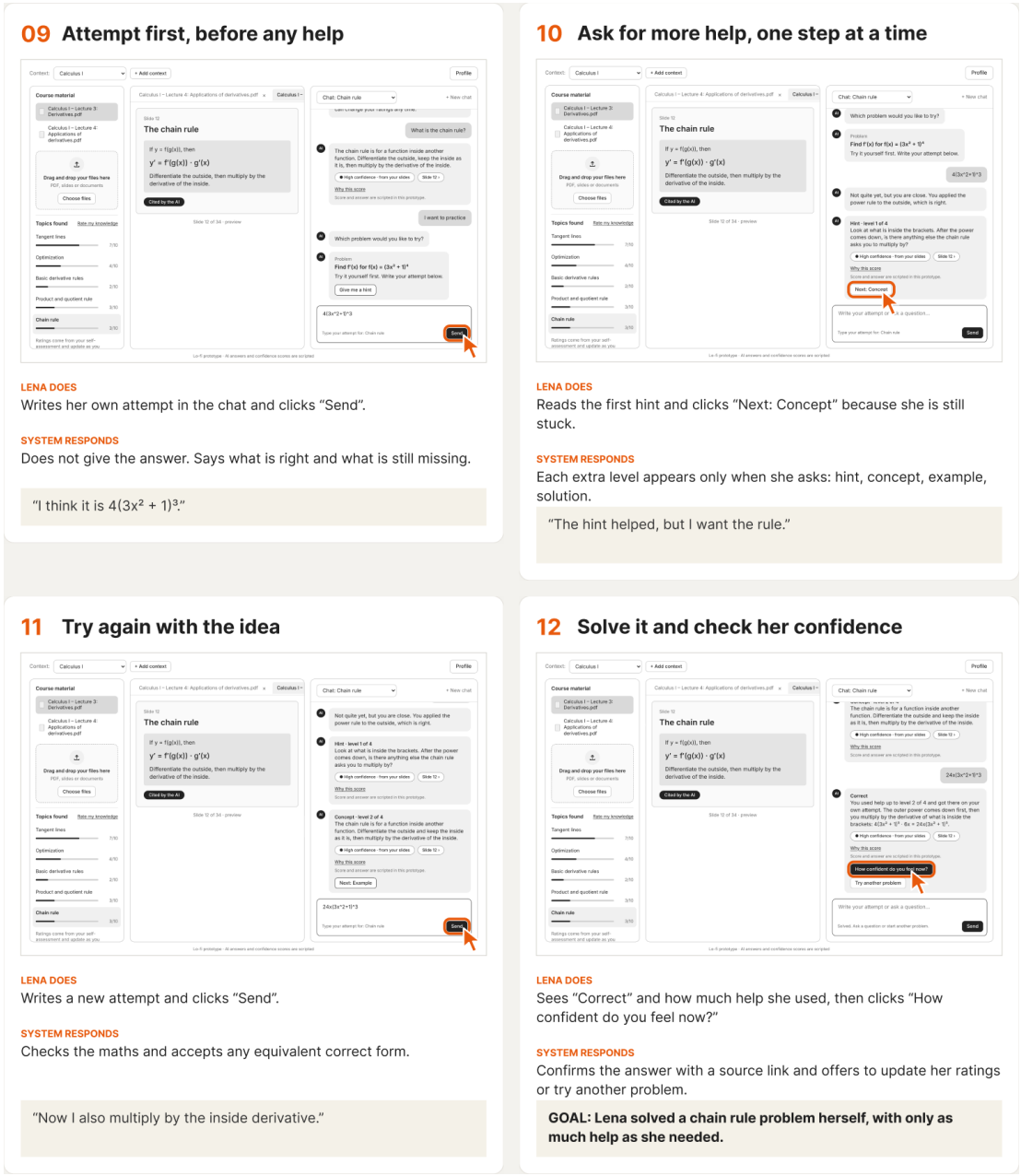


Figure 15: Storyboard – Idea B, panels 09–12. Lena submits an attempt; feedback identifies the missing inner derivative. She requests a hint and then a concept explanation, tries again and reaches a correct answer. The last screen offers a confidence update or another problem.

How the task is accomplished. Lena controls the amount of support and continues from her own attempt. The depicted route stops after the concept explanation; further help can continue to a similar example and a solution. AI replies and confidence scores are scripted for lo-fi testing. The scenario demonstrates the intended interaction, not tested AI accuracy or a peer-testing result.